



Project 3 NLP: Business Case: Automated Customer Reviews

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Project Objectives

Sentiment Analysis -Automatically classify product reviews into Negative/Neutral/Positive.

Product Clustering -Discover natural product categories from unstructured names.

AI Summarization -Generate executive summaries of product performance.



Approach Overview

1. BERT Sentiment Classification

- Fine-tune pretrained model on labeled reviews

2. Semantic Clustering

- Sentence-BERT embeddings + K-Means

3. Generative Summarization

- GPT-3.5 analysis of ratings/complaints

Dataset



- **Dataset: consumer-reviews-of-amazon-product**

- First Dataset: 5000 record
- Second Dataset: 28,000 record



- **Columns 24 in total:**

- **name:** product name
- **review.title**
- **review.text:** Full review content
- **review.rating:** number of stars 1-5
- **primaryCategories:** main category
- **Categories:** all possible categories



Classification

Classification Preprocess



1

Convert numerical star ratings into sentiment classes:

- Negative (0): Ratings ≤ 2 stars
- Neutral (1): Exactly 3 stars
- Positive (2): Ratings ≥ 4 stars

2

Dataset Balancing:

- Merge datasets:
 - All negative/neutral samples from both datasets
 - Randomly sample 1,700 positive reviews from reviews_df.

Data (1): reviews_df

labels	
2	4686
1	197
0	117

Data(2): reviews_df28

labels	
2	25545
0	1581
1	1206

Merged Data

labels	
2	1700
0	1698
1	1403

Classification Preprocess



3

Train-Test Split

- Split before resampling to avoid data leakage.
- Split Ratio: 80% train, 20% test.
- Columns Kept: reviews.text and labels.

Trainset before sampling

```
labels
2      1381
0      1332
1      1127
```

Testset

```
labels
0      366
2      319
1      276
```

4

Oversampling

Oversample (negative/neutral) to match the majority class count (1,381 each)

Trainset after sampling

```
labels
0      1381
1      1381
2      1381
```

5

Tokenizer: BERT's bert-base-uncased

- Truncate/pad texts to 256 tokens.
- Add attention masks to ignore padding.
- Convert datasets to PyTorch format.



Classification with Pre-trained BERT

1 Model Setup

- Used bert-base-uncased, pre-trained and fine-tuned for 3-class sentiment classification
- Customized Trainer class to:
 - Track training and validation loss separately
 - Use manual cross-entropy loss calculation for better transparency

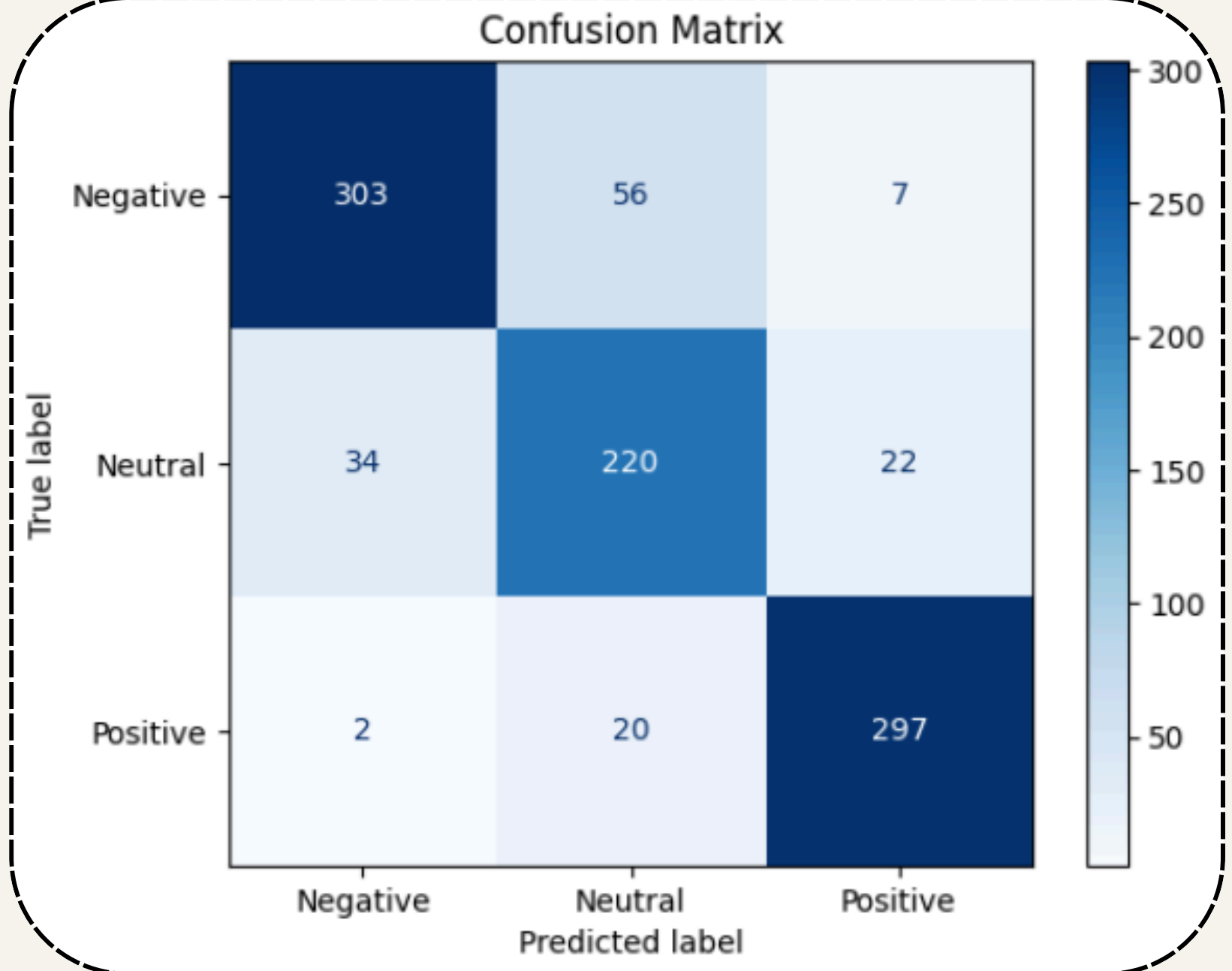
2 Training Configuration

- **TrainingArguments** set for:
 - Epochs:5
 - Batch size: 16
 - Learning rate: $2e-5$
 - Weight decay: 0.01

Classification Results



	precision	recall	f1-score	support
Negative	0.89	0.83	0.86	366
Neutral	0.74	0.80	0.77	276
Positive	0.91	0.93	0.92	319
accuracy			0.85	961
macro avg	0.85	0.85	0.85	961
weighted avg	0.86	0.85	0.85	961





Clustering & Summarizing

Clustering Preprocess



1

Text Cleaning

- Lowercasing & punctuation removal
- Removes stopwords (e.g., the, and)
- Enhances focus on core product terms

2

Deduplication

- Removes duplicate product names after cleaning
- Ensures true product diversity in clusters

3

Semantic Embeddings

- Model: all-MiniLM-L6-v2 (SentenceTransformer)
- Converts names to 384-D vectors
- Captures context & synonyms (e.g., "TV" $\approx$ "Television")

K-Means Clustering for Product Categorization



1

Apply K-Means Algorithm

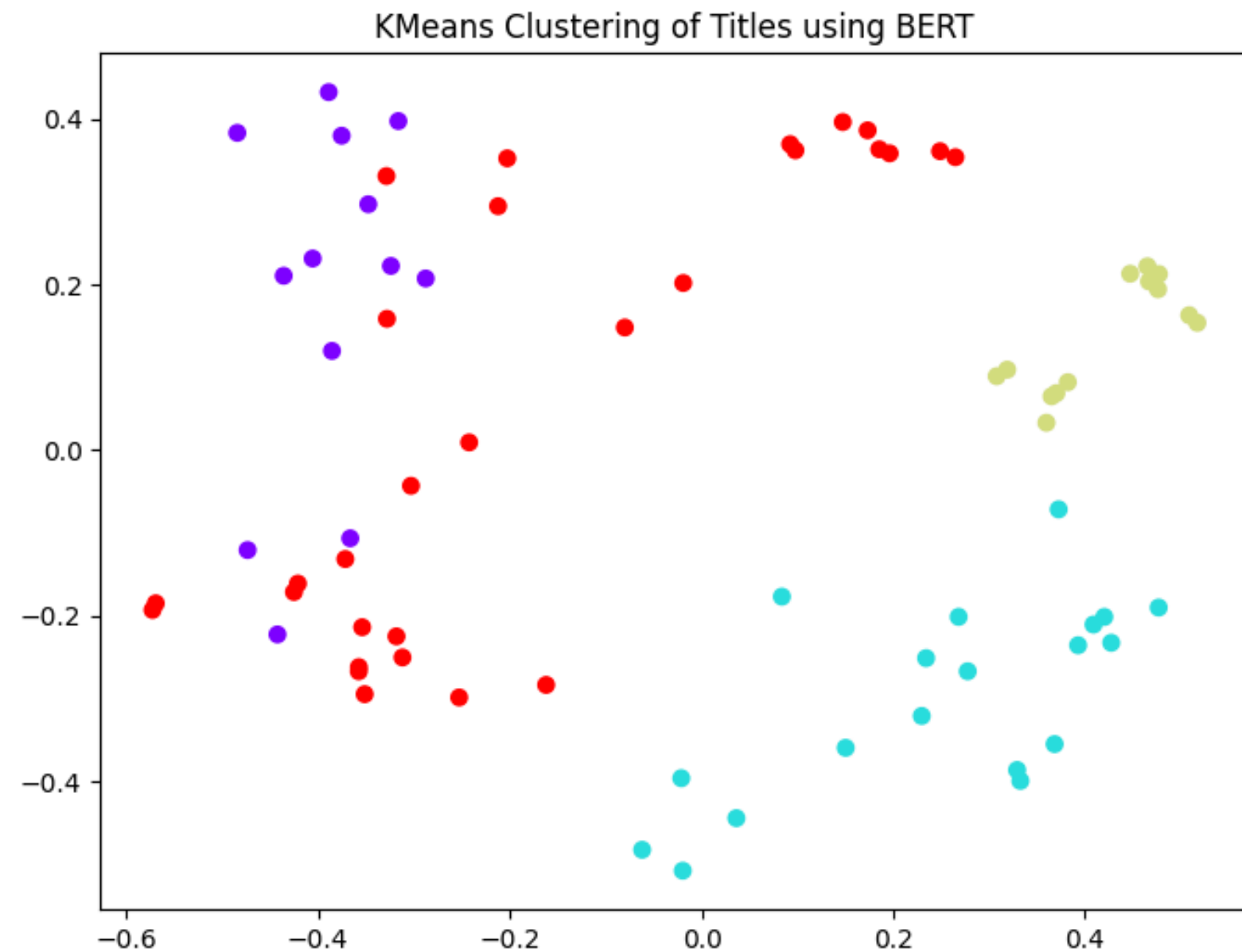
- Chose $K = 4$ for meta-categories
- Used Sentence-BERT embeddings (384D)
- Algorithm groups semantically similar product names

2

Visualize the Clusters

- Removes duplicate product names after cleaning
- Ensures true product diversity in clusters

Clustering Results



Clusters

- Amazon Smart Home & Echo Devices
- "Kindle & E-Reader Products
- Amazon Fire Tablets
- Amazon Fire TV & Electronics Basics

Summarize Reviews Using Generative AI



1 Product Performance Evaluation

- Grouped products by category
- Calculated average ratings
- Selected:
 - 🏆 Top 3 Best-Rated Products
 - ⚠️ Worst-Rated Product in Each Category

2 Complaint Analysis

- Extracted 1-2 star reviews
- Combined titles + review text
- Identified Top 3 Frequent Complaints per product

3 Automated Content Generation with OpenAI

- Structured prompts included:
 - 🏆 Product rankings and average ratings
 - 💡 Key differentiators
 - 🔄 Common complaints
 - ⚠️ Warning about the worst product
- Used OpenAI API to generate natural summaries per category

Overall Result



Classification with Pre-trained BERT

- Accuracy: 0.85
- Macro Avg (F1): 0.85
- Weighted Avg (F1): 0.85

Clustering Product Names with Sentence Transformers

- Model: all-MiniLM-L6-v2 (384-D embeddings)
- Clustering Technique: K-Means (k = 4)

Generative Summarization with OpenAI ChatGPT 3.5

- Automated blog-style content per category
 - ★ Top 3 best products
 - 😡 Key complaints from negative reviews
 - ⚠ Worst product warnings

Challenges:

- Dealing with many new concepts at the same time
- Using Imbalanced data

LAYLA 🕶️



Thank you for listening

ANY QUESTIONS?

LAYLA 🕶️



DEMO Time!



Project Repo:

<https://github.com/LaylaZx/Project-NLP-Business-Case-Automated-Customer-Reviews>